

- 18** A stationary sound wave is set up in a tube such that in an interval of 1.0 s, the displacement of particles at the antinodes swap directions 6000 times. The velocity of the sound wave in the tube is 300 m s^{-1} .

What is the wavelength of the sound wave and at which displacement position will there be a greatest variation of pressure?

	wavelength	greatest pressure variation at
A	10 cm	displacement antinode
B	10 cm	displacement node
C	20 cm	displacement antinode
D	20 cm	displacement node